

Practicle-3(A).

Aim: To create and implement a Java Point class using private fields, constructors, and method overriding, including toString(), equals(), and hashCode(), and to identify and count distinct points by comparing their coordinates.

Code Screenshot:

A) Point.java

```
import java.util.Objects;

public class Point {
    private int x;
    private int y;

    Point(int x, int y) {
        this.x = x;
        this.y = y;
    }

    public String toString() {
        return "(" + x + ", " + y + ")";
    }

    @Override
    public boolean equals(Object obj) {
        Point p = (Point) obj;
        return x == p.x && y == p.y;
    }

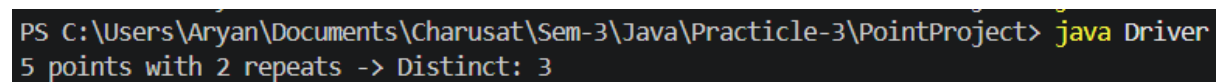
    @Override
    public int hashCode() {
        return Objects.hash(x, y);
    }
}
```

B) Driver.java



```
public class Driver {  
    Run | Debug | Run main | Debug main  
    public static void main(String[] args) {  
        Point[] points = {  
            new Point(x: 1, y: 2),  
            new Point(x: 3, y: 4),  
            new Point(x: 1, y: 2),  
            new Point(x: 5, y: 6),  
            new Point(x: 3, y: 4)  
        };  
  
        int count = 0;  
  
        for (int i = 0; i < points.length; i++) {  
            boolean found = false;  
  
            for (int j = 0; j < i; j++) {  
                if (points[i].equals(points[j])) {  
                    found = true;  
                    break;  
                }  
            }  
  
            if (!found) {  
                count++;  
            }  
        }  
        System.out.println(points.length + " points with " + (points.length - count) + " repeats -> " + "Distinct: " + count);  
    }  
}
```

Output Screenshot:



```
PS C:\Users\Aryan\Documents\Charusat\Sem-3\Java\Practicle-3\PointProject> java Driver  
5 points with 2 repeats -> Distinct: 3
```

Practicle-3(B)

Aim: To create and implement a Java Card class using encapsulation, constructors, and method overriding with toString(), equals(), and hashCode(), and to identify duplicate cards by comparing their rank and suit.

Code Screenshot:

A) Card.java

```
import java.util.Objects;

public class Card {
    private String suit;
    private String rank;

    public Card(String suit, String rank) {
        this.suit = suit;
        this.rank = rank;
    }

    @Override
    public String toString() {
        return suit + " of " + rank;
    }

    @Override
    public boolean equals(Object obj) {
        Card c = (Card) obj;
        return suit == c.suit && rank == c.rank;
    }

    @Override
    public int hashCode() {
        return Objects.hash(suit, rank);
    }
}
```

B) Driver.java

```

public class Driver {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {

        Card[] cards = new Card[5];

        cards[0] = new Card(suit: "Ace", rank: "Spades");
        cards[1] = new Card(suit: "King", rank: "Hearts");
        cards[2] = new Card(suit: "Queen", rank: "Diamonds");
        cards[3] = new Card(suit: "Ace", rank: "Spades");

        for (int i = 0; i < cards.length; i++) {

            for (int j = 0; j < i; j++) {

                if (cards[i].equals(cards[j])) {

                    System.out.println("a second " + cards[i] + "->" + " Duplicate found: " + cards[i]);
                    return;
                }
            }
        }
    }
}

```

Output Screenshot:

```

PS C:\Users\Aryan\Documents\Charusat\Sem-3\Java\Practicle-3\CardProject> java Driver
a second Ace of Spades -> Duplicate found: Ace of Spades

```

- **Question-Answer: -**

1. **Why must equals() and hashCode() be overridden together?**

- equals() and hashCode(): They must be overridden together so equal objects always have the same hash code, especially in collections like HashSet and HashMap.

2. **What does the default Object.toString() return, and why override it?**

- Default toString(): It returns the class name followed by the object's hash code. We override it to display meaningful object information.

3. **What is the difference between a static nested class and an inner class?**

- Static nested vs inner class: A static nested class does not need an outer-class object, while an inner class is associated with an instance of the outer class.

Learning Outcome:

The learning outcome is to understand and implement equals(), hashCode(), and toString() methods in Java. Students will learn how to compare objects based on their data and identify duplicate objects. They will understand how arrays can be used to store and process objects. They will also learn to check distinct values and handle duplicate

records. This activity improves understanding of object comparison and object-oriented programming.

GitHub Repositories: -

<https://github.com/d26it116/OOP/tree/159f27c0b1a369ac4529973d98446af6a77befb1/Practice-3>